Chapter 10
Carbohydrate Microarray Technology



Abstract Carbohydrate microarrays allow the rapid screening of interactions between glycans and other molecules, such as proteins, other carbohydrates, or even whole bacteria. It is shown that a great variety of proteins with binding properties for specific glycan epitopes are used for lectin microarrays. Nowadays, microarray methods are also used for direct analysis of live mammalian cell-surface glycome. This chapter gives an introduction to the extensive microarray technology.

Keywords Glycobiosensors · Lectins · Antibodies · Glycan-binding protein (GBP) · Fluorescence · GLAD · GlyMDB · CarbArrayART

As has been shown in Chap. 1, carbohydrates play pivotal roles in cell functions. For example, cell–cell recognition, cell signaling, and cell trafficking are mediated through interactions of carbohydrates with other biomolecules, such as (glyco)proteins, (glyco)peptides, and other carbohydrates. A study of the presence or absence of binding between specific molecules can reveal important physiological and structural information for both the binding molecule and the ligand alike.

The carbohydrate microarray technology (also known as glycobiosensors) provides a high-throughput method that enables screening of a multitude of carbohydrates against a multitude of binding partners to determine and evaluate their interactions [1–5]. For a long time, microarray technologies have been used in DNA and RNA research, but it has been only less than two decades ago that this technique was applied in glycoscience. In recent years, because of the increasing development of high-throughput technologies such as lectin arrays and glycan arrays, several databases have been established to store and share the glycan–protein interaction data (see Chap. 13).

In a typical glycan microarray experiment, many different glycan structures are immobilized in a spatially defined and miniaturized fashion on a solid surface, such as a glass slide, for investigation of their interrogation with carbohydrate-binding compounds. The carbohydrates used for immobilization can be either chemically synthesized or isolated from natural sources. Compared to other analytical methods, the amounts of material needed for printing microarrays are very small (femtomole) and use minuscule volumes (picoL). This enables high-density printing of over 1000 spots of different carbohydrates (i.e., potential unique carbohydrate structures)

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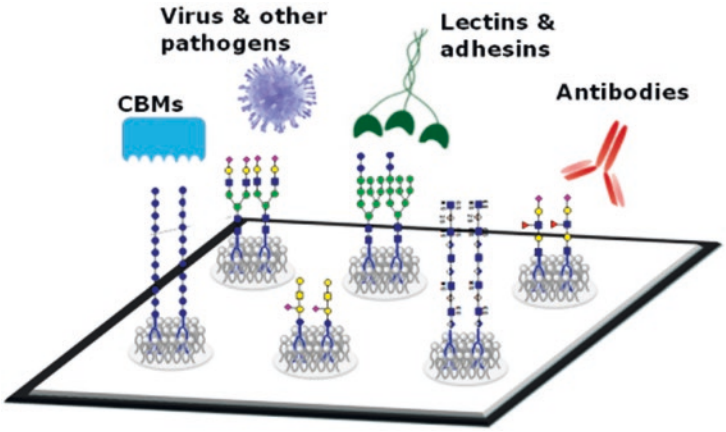


Fig. 10.1 Schematic presentation of a carbohydrate microarray, illustrating possible binding interactions with different ligands (Creative Proteomics, USA)

per microarray the size of a microscope slide. Such a slide can accommodate 100 or more probing experiments at once, making it possible to monitor multiple molecular interactions with biomolecules where glycans are involved (Fig. 10.1) [6, 7]. Additionally, by compartmentalizing the microarray slides with hydrophobic compounds such as Teflon into "multi-well" microarray slides, the volumes of typically precious reagents used for probing are also strongly minimized, enabling efficient, multiplexed, and cost-effective experiments.

The National Consortium for Functional Glycomics (CFG) (https://ncfg.hms.harvard.edu/microarrays) has a microarray resource with the aim to develop carbohydrate (micro)arrays as innovative tools to map highly specific interactions between carbohydrate and protein partners [8–10].

In very general terms, select carbohydrates (mono/oligo/polysaccharides) are bound (printing/blocking) to the microarray surface. This can be obtained by non-covalent interactions through conjugation to hydrophobic molecules and immobilization on nitrocellulose membranes or covalent attachment mediated by functional groups on the oligosaccharides and the microarray surface [9, 11]. The preparation of glycan microarrays requires that the glycans are derivatized with functional groups, such as an amino group. Polysaccharides can be immobilized on a variety of microarray surfaces (e.g., nitrocellulose membranes, and polystyrene) without a need for chemical functionalization because of their size that multiplies the number of nonspecific interactions enabling their adhesion.

Mono- and oligosaccharides do not have this advantage and typically need to be functionalized synthetically to be attached to the solid surface. This can be achieved either by using (a) a functionalized surface and a non-modified carbohydrate moiety or (b) a functionalized surface, and a functionalized carbohydrate.

A general strategy to construct glycan microarrays is the covalent bond formation of glycans containing reactive amine groups (e.g., aminoethyl) at the reducing

termini. Using aldehyde-modified glass microscope slides, the primary amines of the modified sugars react with the aldehyde groups to form a Schiff base linkage. Optionally, reducing oligosaccharides may be reductively aminated using an amino lipid, thereby introducing an amphipathic property to the oligosaccharide. The hydrophobic lipid tail facilitates immobilization on silica gel surfaces, plastic microtiter plates, membranes, and microscope slides [12, 13]. Several covalent attachment techniques, for instance using carrier proteins or biotin/streptavidin linkage, can be found in the carbohydrate microarray literature [5].

The immobilized glycans are deposited on slides in a spatially discrete pattern using automated arraying robots, usually at different concentrations in several replicates. Then, the slides are interrogated or incubated with (fluorophore-labeled) compounds that may bind glycans, including cells, (recombinant) proteins, lectins, antibodies, toxins, serum, viruses, and bacteria (Fig. 10.1). Subsequently, binding is detected by direct or indirect fluorescence approaches, which provide an image of fluorescent spots that may be identified to coincide with deposited glycans. The data intensity of fluorescence is measured (scanned) and graphed (quantifying the fluorescence intensity of the bound compound) versus glycan number on the microarray. In this way, glycan microarrays can be used to discover functional glycans, characterize glycan-processing enzymes, and detect pathogens for diagnosis. Binding preferences, inhibition of interactions, enzyme activities, and structure–function relationships can be determined.

Recently, it was found that the fluorescent 2-AA tag of glycans (see Sect. 5.4.2), which is helpful for detection during HPLC, can serve as a linker by using the carboxylic acid functional group for selective amidation with various amines. This opens a method enabling facile immobilization of glycans to solid phases for the preparation of glycan microarrays [14].

In contrast to immobilizing carbohydrates, a multitude of specific lectins (Table 10.1) can be immobilized on the slide to produce an array to assess the specificity of binding properties to carbohydrates [15–18]. Each lectin possesses binding activity toward different glycan epitopes and incubation of the array with a fluorescently labeled (e.g., glycoprotein) sample reveals which of these epitopes are present. These lectin microarrays can be used for rapid analysis of protein glycosylation of therapeutic proteins, although they will not provide a complete structure assignment but a fingerprint pattern [19]. On lectin microarrays, also whole cells and intact extracellular vesicles can be used [20].

From the list of lectins in Table 10.1, there are five that are found in pretty much everybody's lectin toolbox, being Con A, RCA, WGA, PNA, and SNA, because they are specific for binding mainly to one terminal monosaccharide. Most of the other lectins have more complex binding motifs (multivalent), which need careful consideration to avoid wrong conclusions.

Clearly, microarrays are outstanding tools for the study of cellular glycosylation patterns and interactions of biomolecules with carbohydrates. Glycan analogs, non-natural oligosaccharides, glyco-mimetics, synthetic glycoconjugates, and inhibitors can serve as probes [21]. In particular, the lectin array-based affinity test is becoming an important tool for isomeric glycan analysis [22, 23]. As such, lectin

Table 10.1 Lectins frequently used for carbohydrate identification in microarray and in affinity separation methods (terminal A in abbreviation usually stands for Agglutinin)

Lectin	Source	Carbohydrate specificity
AAA	*Anguilla anguilla*	Fuc(α1→3/6)
AAL	*Aleuria aurantia*	Fuc(α1→6)GlcNAc, Fuc(α1→3)[Gal(β1→4)]GlcNAc
ABA	*Agaricus bisporus*	Gal(β1→3)GalNAc
Con A	*Canavalia ensiformis*	Man(α1→6)[Man(α1→3)]Man (biantennary)
DSL	*Datura stramonium*	GlcNAc(β1→4)GlcNAc, Gal(β1→4)GlcNAc
DBA	*Dolichos biflorus*	GalNAc(α1→3)GalNAc, Terminal α-GalNAc
ECA	*Erythrina cristagalli*	Gal(β1→4)GlcNAc, Terminal GalNAc
GNA	*Galanthus nivalis*	Man(α1→3)Man
GNL	*Galanthus nivalis*	Man(α1→3)Man, High-Man N-glycans
GSL	*Griffonia simplicifolia*	Tri/tetra-antennary N-glycans with terminal GlcNAc, Terminal Gal(α1→3)
HHL	*Hippeastrum* hybrid	High-Man N-glycans, Man(α1→3)Man, Man(α1→6)Man
HAA	*Helix aspersa*	GalNAcα/β
HPA	*Helix pomatia*	GalNAcα/β
LCA	*Lens culinaris*	α-Glc, α-Man in N-glycans with Fuc(α1→6)GlcNAc
LTA	*Lotus tetragonolobus*	Fuc(α1→2)Gal(β1→4)[Fuc(α1→3)]GlcNAc
MAA	*Maackia amurensis*	Neu5Ac(α2→3)Gal(β1→4)GlcNAc
MAL-I	*Maackia amurensis*	Neu5Ac(α2→3)Gal(β1→4)GlcNAc
MAL-II	*Maackia amurensis*	Neu5Ac(α2→3)Gal(β1→3)GlcNAc, Neu5Ac(α2→6)GalNAc
MPA	*Maclura pomifera*	Galα, GalNAcα
NPA	*Narcissus pseudonarcissus*	Terminal Manα
NPL	*Narcissus pseudonarcissus*	High-man N-glycans, Man(α1→6)Man
PHA-E	*Phaseolus vulgaris*	Gal(β1–4)GlcNAc(β1–2)Man with bisecting GlcNAc
PHA-L	*Phaseolus vulgaris*	Tri/tetra-antennary complex N-glycans with terminal Gal
PhoSL	*Pholiota squarrosa*	Fuc(α1→6)GlcNAc
PNA	*Arachis hypogaea*	Gal(β1→3)GalNAc, Gal(β1→4)Glc
PSA	*Pisum sativum*	Manα, Glcα
RCA	*Ricinus communis*	Gal(β1→4)GlcNAc
SBA	Glycine max	GalNAc(α1→3)Gal, Terminal GalNAc
SNA	*Sambucus nigra*	Neu5Ac(α2→6)Gal(NAc), Galβ
TL	Tomato lectin	Poly-*N*-acetyllactosamine
VVA	*Vicia villosa*	Terminal GalNAcα/β
WFA	*Wisturia floribunda*	GalNAc(β1→4)-terminated glycans
WGA	*Triticum vulgaris*	GlcNAc(β1→4)GlcNAc, chitin oligos, Neu5Ac
UEA-I	*Ulex europaeus*	Fuc(α1→2)Gal(β1→4)GlcNAc(β1→

microarrays are a sensitive, rapid, and high-throughput profiling method, which enable the analysis of lectin–glycan interactions in a simultaneous manner. In general, labeled glycoproteins applied on the lectin array side, which are bound, are detected using an array scanner. Lectin arrays are used in the identification of desired glycosylation patterns during clone screening and selection during process development. Furthermore, lectin arrays are excellent tools in biomedical research, useful for pathogen detection, biomarker discovery, and bacterial tropism.

A lectin array-based method can be used for rapid analysis of glycosylation profiles of glycoproteins and mammalian cell surfaces [24]. The method is based on binding an intact glycoprotein to the arrayed lectins, resulting in a characteristic fingerprint that is highly sensitive to changes in glycan composition. A large number of lectins, each with its specific recognition pattern, ensures high sensitivity to changes in the glycan pattern. In fact, the lectin microarray has been recognized as a unique method to analyze glycosylation features of diverse glycoproteins, which include those of crude cell lysates, sera, and bacteria. A set of proprietary algorithms can automatically interpret the fingerprint signals to provide a comprehensive glycan profile output [15]. Sophisticated tools, such as Glycan Array Dashboard (GLAD) and Glycan Microarray Database (GlyMDB), are accessible nowadays [25, 26].

However, there are some conceptual challenges with microarray glycomics because the structural specifics of lectin-epitope binding are not completely understood. There is a strong overlap in carbohydrate binding for many lectins. The detailed characterization of many lectin specificities is still in progress. Most lectins are multimeric and represent avidities rather than one-to-one events, such as neighboring residues. Also, three-dimensional (3D) features of glycans influence the recognition by glycan-binding proteins (GBPs). Thus, the absolute selectivity of a lectin for a class of carbohydrates cannot always be guaranteed. Furthermore, it has to be noted that most lectins seem more effective at the level of intact glycoproteins rather than glycopeptides. The scanning of a microarray, with detection of binding, usually read as fluorescent intensity, the information gathered must be handled with care concerning the quantitative determination. The level of information that a microarray can typically provide is (a) no binding, (b) low binding, (c) medium binding, or (d) high binding. Therefore, treating the data as categorical instead of continuous is generally a more reasonable approach to microarray data analysis.

Nevertheless, a major advantage of lectin arrays is the unbiased glycan analysis over all classes of carbohydrates on intact cells or vesicles, which do not require exhaustive sample preparation. Glycoarrays open the way for a wide variety of biomedical applications, such as the identification of novel sugar markers related to early-stage diseases and the detection and diagnosis of pathogenic infections. Furthermore, the system is a suitable means for quality control of various glycoprotein therapeutics products [24].

A wide array of standard oligosaccharides as pyridylaminated (PA) derivatives is commercially available as a valuable tool for the systematic and detailed analyses of structures and functions of glycans. In recent years, this has led to an explosive growth of data from carbohydrate microarray experiments, coming from multiple

research laboratories, each employing their own proprietary technology for spotting the array. However, the often radically different approaches of the spotting of the arrays can change the binding affinities observed and cause problems in comparative analysis and evaluation of the data.

Commercial microarrays and advanced commercial instruments/scanners with dedicated software (e.g., SensoSpot^{R} Microarray Analyzers, Array-Pro Analyzer, and GlycoStation) are now becoming available, designed for routine microarray analyses, using fluorescence and colorimetric detection (GP Biosciences, Media Cybernetics, GE Healthcare).

Microarrays can also be prepared from glycopeptides/proteins or glycolipids. Also, mass spectrometry techniques are employed as label-free detection methods in glycan microarrays. The advances in chemical synthesis over the past decade have now made it possible to produce targeted glycans. Since facile methods for producing compound libraries are quickly improving, the use of carbohydrate arrays will rapidly increase.

Despite the several advantages, lectin arrays do not provide detailed information on sialic acid speciation and oligosaccharide sequences and linkages.

The US Consortium for Functional Glycomics (CFG) (http://www.functionalglycomics.org) supported the development of a robot-produced, publicly available microtiter-based glycan array. The currently used printed mammalian glycan microarray format (version 4.1) comprises >450 synthetic and natural glycan sequences representing major glycan structures of glycoproteins and glycolipids [27]. In 2008, a pathogen glycan array was also made available for screening, containing 96 polysaccharides derived from Gram-negative bacteria [28]. Furthermore, there is a role for lectin microarrays in cancer diagnosis [29]. CarbArrayART (Carbohydrate microarray Analysis and Reporting Tool) is a freely available software, based on the GRITS Toolbox [30], for storage, processing, and presentation of carbohydrate microarray data. Since 2020, the GLYCAN GlyMDB covers over 5000 microarray samples information (http://www.glycanstructure.org/glymdb/).

It is evident that for functional studies glycan microarray has become an indispensable technology for screening carbohydrate interactions in biological research and medicine [31–33]. However, implementation of the carbohydrate microarray technique is often hampered by the thought that the method requires an extensive degree of specialized training and a high level of experience. Some methodologies involved in the fabrication of microarrays indeed require moderate expertise in chemistry, starting from modifying carbohydrates through attaching them to solid surfaces, but detailed protocols can make microarray technology accessible even to non-experts through the use of protocols comparable to methodologies as old as Western blot analysis. Studying glycan interactions will be extremely important to find biomarkers for disease.

Here, only a short introduction to glycan microarrays could be provided. The topic is so extensive that it requires a book in its own right to explain all aspects and discuss the many protocols, including array fabrication and elaborate screening procedures. Researchers, interested in starting up microarray experiments, are referred to books, like "Microarrays" [34], "Microarray Technology" [35], and "Carbohydrate Microarrays: Methods and Protocols" [36].

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